



Lingjing Platform

Simulators



AirSim



CARLA



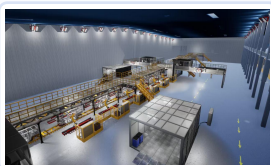
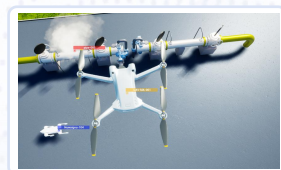
MuJoCo



Discrete
Time Sim.



Unreal



Adapter



Vision



Status



Agents



UAV



Vehicle



Robot



Vessel



Command

Data & Services



Scene Library
(3D Assets / Maps)



ModelZoo
(Pretrained Models)



Knowledge Base
(Ontology/Rules)



API & SDK
(REST/gRPC)



Monitoring
& Visualization



Perceive



Think



Act



Communicate



Step: Simulation Advance

Analysis & Evaluation

Metrics



Visualization

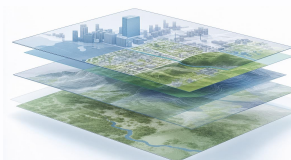


Report



Urban Physics Engine

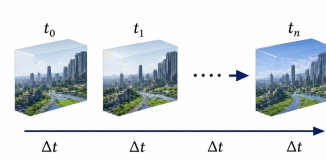
Environment



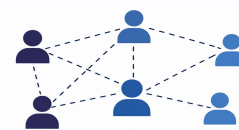
Physics Models

$$x_{t+1} = f(x_t, u_t, w_t)$$
$$y_t = h(x_t, v_t)$$

Temporal Evolution



Multi-Agent Interaction



Uncertainty & Noise



Scalability

